A COMPREHENSIVE STUDY OF REGULARISATION TECHNIQUES IN MACHINE LEARNING

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1. Introduction

When exploring machine learning, a model with high training accuracy does not always perform well in real-world scenarios. A model can "memorise" training data—capturing noise and random fluctuations rather than meaningful patterns—a phenomenon known as overfitting. Regularisation addresses this by adding constraints to prevent excessive complexity, ensuring models focus on important patterns and generalise well to new data. This has shifted my focus from simply maximising accuracy to building models that are reliable and robust.

2. Understanding the Core Problem

2.1 Overfitting

Overfitting occurs when a model learns training data too well, capturing noise alongside real patterns. It achieves high training accuracy but performs poorly on unseen data. This is like a student who memorises answers—they struggle when questions are phrased differently.

2.2 Underfitting

Underfitting occurs when a model is too simple to capture the underlying data structure, performing poorly on both training and test data—like a student who has not studied enough to answer even basic questions.

2.3 Bias-Variance Tradeoff

Bias refers to errors from oversimplified assumptions (causing underfitting); variance refers to sensitivity to small data changes (causing overfitting). The goal is a balance where the model complex enough to learn patterns but simple enough to generalise well.

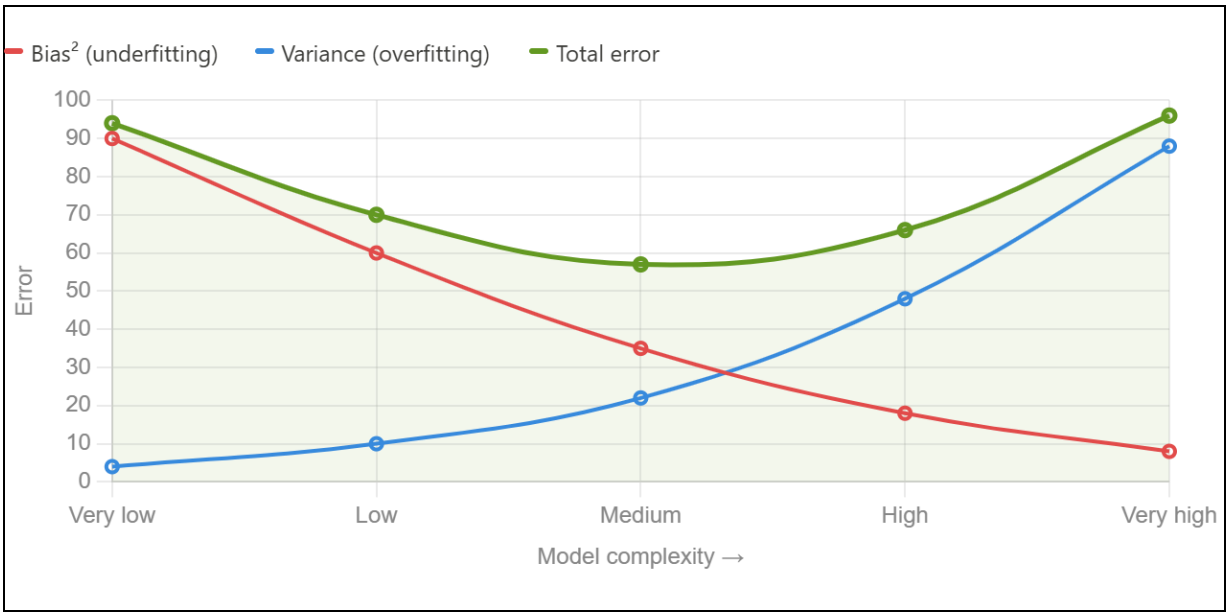

Fig 1 The bias-variance tradeoff

2.4 Real-World Example: House Price Prediction

An overfitted house price model may associate high prices with irrelevant features like door colour—accurate on training data but unreliable in reality. An underfitted model using only one feature (e.g., room count) produces inaccurate predictions across all data. Both extremes highlight why controlling model complexity is essential.

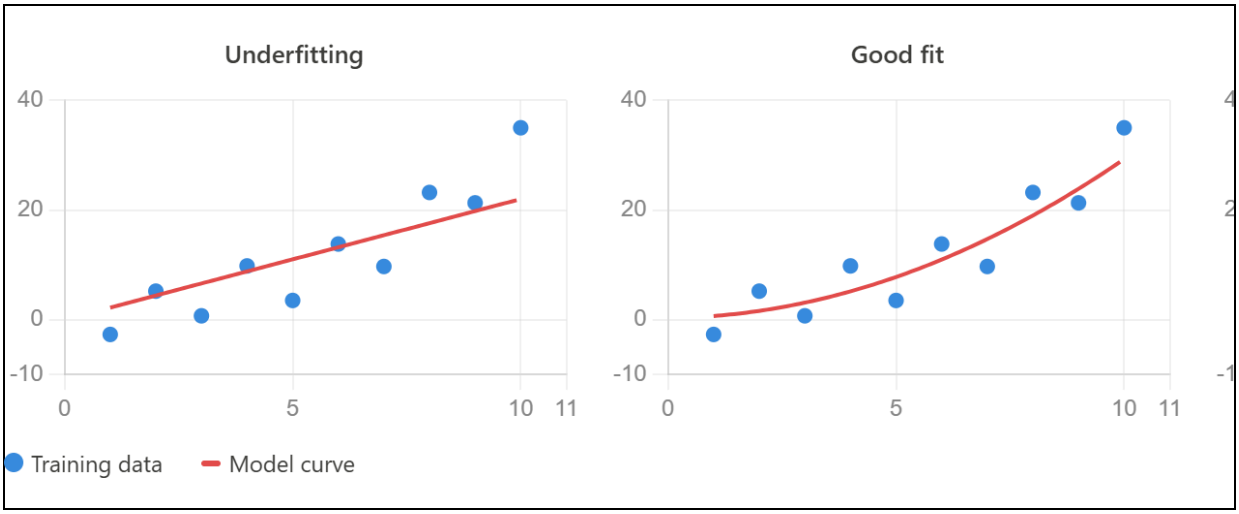

Fig 2 Two fitting scenarios on the same noisy dataset

3. What is Regularisation?

Regularisation guides the model to learn general patterns rather than memorise data—like a teacher encouraging a student to understand concepts rather than memorise answers. It adds a penalty for large coefficient values, restricting how flexible the model becomes. This results in smoother, simpler models that are less sensitive to data fluctuations and perform better on unseen data. Regularisation reduces variance without significantly increasing bias, achieving the optimal bias-variance balance.

4. Types of Regularisation Techniques

4.1 L1 Regularisation (Lasso)

Lasso (Least Absolute Shrinkage and Selection Operator) adds a penalty based on the absolute values of coefficients:

$$\boldsymbol{Loss = RSS + \lambda \Sigma |w_j|}$$

A key property is featuring selection: L1 can shrink some coefficients to exactly zero, effectively removing less important features. This is especially useful for high-dimensional datasets. For example, in a healthcare model predicting disease risk, L1 regularisation can automatically prioritise critical factors like genetic markers and blood pressure while eliminating irrelevant variables, making the model simpler and more interpretable for clinicians.

4.2 L2 Regularisation (Ridge)

Ridge Regression adds a penalty based on squared coefficient values:

$$\boldsymbol{Loss = RSS + \lambda \Sigma w_j^2}$$

Unlike L1, Ridge shrinks all coefficients but never reaches zero, ensuring every feature retains some influence. This weight-shrinking effect prevents extreme values and improves model stability. In stock price prediction, where factors like historical prices, market trends, and economic indicators all contribute, L2 ensures a balanced contribution without over-reliance on any single variable, leading to more stable predictions.

4.3 Elastic Net Regularisation

Elastic Net combines both L1 and L2 penalties:

$$Loss = RSS + \lambda_1 \Sigma|w_j| + \lambda_2 \Sigma w_j^2$$

It is preferred when datasets have many correlated features—L1 alone may arbitrarily drop useful features, while L2 alone may not simplify enough. For example, in predicting e-commerce customer behaviour, features like income, browsing history, and purchase frequency are often correlated. Elastic Net selects the most relevant feature groups while controlling their impact, improving both accuracy and interpretability.

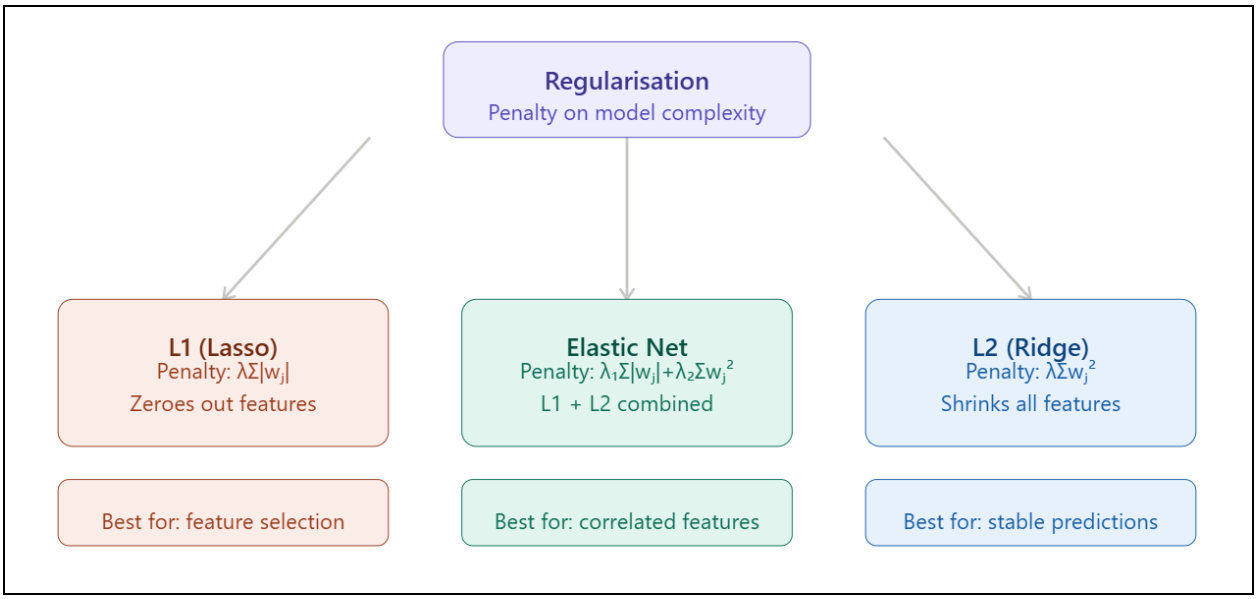

Fig 3 Overview of the three main regularisation techniques

5. Mathematical Overview

Without regularisation, models minimise the Residual Sum of Squares (RSS):

$$Loss = \Sigma(y_i - \hat{y}_i)^2$$

Focusing solely on minimising error risks overfitting. Regularisation adds a penalty term to the loss function, penalising large coefficients. The model now balances two objectives: minimise prediction error and keep coefficients small. As coefficients grow, the penalty increases the loss, forcing the model to stay simple and avoid capturing noise.

6. Hyperparameter Tuning (Lambda λ)

Lambda (λ) controls the strength of regularisation. It is set before training and directly influences model complexity:

- **Small λ (near zero):** minimal regularisation, risk of overfitting
- **Large λ:** strong regularisation, risk of underfitting

Finding the optimal λ is done through hyperparameter tuning: trial and error, cross-validation (evaluating generalisation on a held-out set), or grid search (systematically testing a range of values). The best λ minimises validation error while avoiding both extremes.

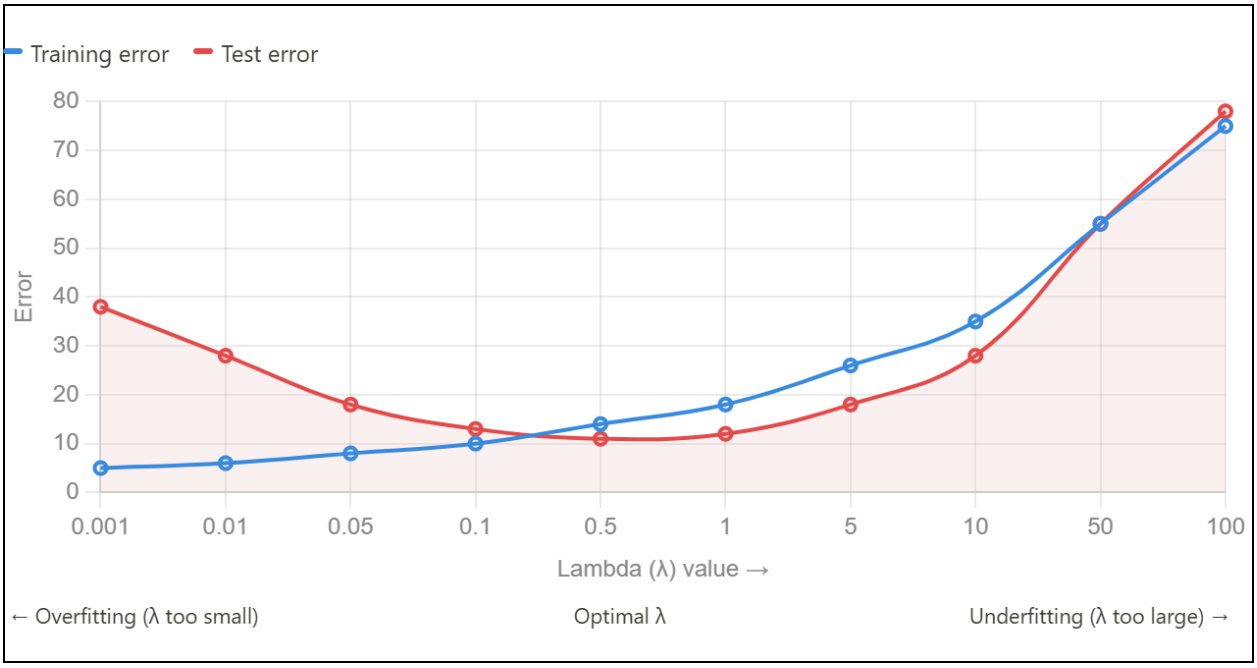

Fig 4 The effect of the regularisation parameter λ on training and test error

7. Implementation (Python / Scikit-learn)

The following code compares Linear Regression, Ridge (L2), and Lasso (L1) on a simulated house price dataset:

```
# Import libraries
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.metrics import mean_squared_error
X = np.random.rand(100, 3)
y = X @ np.array([3, 5, 2]) + np.random.randn(100) * 0.5
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
 lr = LinearRegression().fit(X_train, y_train)
ridge = Ridge(alpha=1.0).fit(X_train, y_train)
lasso = Lasso(alpha=0.1).fit(X_train, y_train)
 print('Linear MSE:', mean_squared_error(y_test, lr.predict(X_test)))
print('Ridge MSE:', mean_squared_error(y_test, ridge.predict(X_test)))
print('Lasso MSE:', mean_squared_error(y_test, lasso.predict(X_test)))
```

Regularised models typically show slightly higher training error but lower test error, confirming better generalisation. Ridge reduces all coefficient magnitudes for stability; Lasso may zero out some features, simplifying the model.

8. Conclusion

Regularisation is a fundamental principle in machine learning that balances model complexity and generalisation. By applying L1 for feature selection, L2 for stable predictions, or Elastic Net for correlated feature sets—and tuning λ appropriately—practitioners can build models that are robust, interpretable, and effective in real-world applications. Understanding regularisation has shifted my focus from simply maximising training accuracy to building models that truly generalise well.

9. References

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